

The planes

1. The general equation of plane is

$$ax + by + cz + d = 0$$

2. The general equation of a plane which passes through a point (x_1, y_1, z_1) is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$

3. The general equation of a plane which passes through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) is $(x - x_1)(y - y_2) - (y - y_1)(x - x_2) = k \{ (x - x_1)(z - z_2) - (x - x_2)(z - z_1) \}$

4. The general equation of a plane which passes through the points (x_1, y_1, z_1) , (x_2, y_2, z_2) and (x_3, y_3, z_3) is

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

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5. The equation of a plane which cuts intercepts a, b, c from the axes is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

6. If the points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) and (x_4, y_4, z_4) are coplanar then

$$\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0$$

7. The equation of a plane when the direction cosines of its normal are l, m, n and perpendicular distance p from origin is $lx + my + nz = p$

8. The equation of the plane which passes through the intersection line of the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is $a_1x + b_1y + c_1z + d_1 + k(a_2x + b_2y + c_2z + d_2) = 0$

9. If α be the angle between the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$

$$\text{then } \cos \alpha = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$a_1a_2 + b_1b_2 + c_1c_2 = 0 \rightarrow \text{perpendicular}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \rightarrow \text{parallel}$$

10. The parallel plane of $ax + by + cz + d = 0$ is $ax + by + cz + k = 0$.

11. The equation of plane which passes through the point (x_1, y_1, z_1) and perpendicular to the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$

$$\text{is } \begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0.$$

12. The perpendicular distance from (x_1, y_1, z_1) to the plane $ax+by+cz+d=0$ is $\left| \frac{ax_1+by_1+cz_1+d}{\sqrt{a^2+b^2+c^2}} \right|$

13. The general equation of second degree $ax^2+by^2+cz^2+2fyz+2gzx+2hxy=0$ represent a pair of plane if $abc+2fgh-af^2-bg^2-ch^2=0$.

14. The equations of two bisector planes of the angles between two planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ are

$$\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}} = \pm \frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}$$

15. If $ax^2+by^2+cz^2+2fyz+2gzx+2hxy=0$ represent a pair of plane then the angle between the planes is

$$\cos \theta = \frac{2\sqrt{f^2+g^2+h^2-abc}}{a+b+c}$$

16. The volume of a tetrahedron whose vertices are (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) and (x_4, y_4, z_4) is

$$V = \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix}$$

Problem-1: Find the equation of the plane through the three points $(4, 2, 1)$, $(-1, -2, 2)$, $(0, 4, -5)$.

Solution: The equation of a plane which passes through the points $(4, 2, 1)$, $(-1, -2, 2)$ and $(0, 4, -5)$

$$\text{is } \begin{vmatrix} x & y & z & 1 \\ 4 & 2 & 1 & 1 \\ -1 & -2 & 2 & 1 \\ 0 & 4 & -5 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-4 & y-2 & z-1 & 0 \\ 5 & 4 & -1 & 0 \\ -1 & -6 & 7 & 0 \\ 0 & 4 & -5 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-4 & y-2 & z-1 \\ 5 & 4 & -1 \\ -1 & -6 & 7 \end{vmatrix} = 0$$

$$\Rightarrow 22(x-4) - 34(y-2) - 26(z-1) = 0$$

$$\Rightarrow 11x - 17y - 13z + 3 = 0. \underline{\underline{\text{Ans}}}$$

Problem-2: Show that the points $(-6, 3, 2)$, $(3, -2, 4)$, $(5, 7, 3)$ and $(-13, 17, -1)$ are coplanar.

Solution: Now,
$$\begin{vmatrix} -6 & 3 & 2 & 1 \\ 3 & -2 & 4 & 1 \\ 5 & 7 & 3 & 1 \\ -13 & 17 & -1 & 1 \end{vmatrix} = \begin{vmatrix} -9 & 5 & -2 & 0 \\ -2 & 9 & 1 & 0 \\ 18 & -10 & 4 & 0 \\ -13 & 17 & -1 & 1 \end{vmatrix}$$

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$$= \begin{vmatrix} -9 & 5 & -2 \\ -2 & -9 & 1 \\ 18 & -10 & 4 \end{vmatrix}$$

$$= -9(+36+10) + 5(18+8) - 2(20-162)$$

$$= 0 \text{ (shown)}$$

Problem-3: Find the equation of the plane which passes through the point $(1, -5, -2)$ and whose normal has direction ratio $2, 3, -7$.

Solution: The equation of a plane which passes through a point (x_1, y_1, z_1) is $a(x-x_1) + b(y-y_1) + c(z-z_1) = 0$

$$\Rightarrow 2(x-1) + 3(y+5) - 7(z+2) = 0$$

$$\Rightarrow 2x + 3y - 7z - 1 = 0$$

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Problem-4: Find direction cosines and length of the normal to the plane $9x + 6y - 2z + 7 = 0$.

Solution: Given that, $\frac{9}{\sqrt{9^2+6^2+(-2)^2}}x + \frac{6}{\sqrt{9^2+6^2+(-2)^2}}y$

$$- \frac{2}{\sqrt{9^2+6^2+(-2)^2}}z = - \frac{7}{\sqrt{9^2+6^2+(-2)^2}}$$

$$\Rightarrow \frac{9}{11}x + \frac{6}{11}y - \frac{2}{11}z = - \frac{7}{11}$$

$\therefore$ direction cosines are $\frac{9}{11}, \frac{6}{11}, -\frac{2}{11}$ and length
 $= \left| -\frac{7}{11} \right| = \frac{7}{11}$

problem-5: Find the acute angle between the planes $3x+2y-5z-4=0$ and $2x-3y+5z-8=0$

solution: Let, θ be the angle between the planes $3x+2y-5z-4=0$ and $2x-3y+5z-8=0$

$$\text{we know } \cos \theta = \left| \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}} \right|$$

$$\Rightarrow \cos \theta = \left| \frac{3 \cdot 2 + 2 \cdot (-3) + (-5) \cdot 5}{\sqrt{3^2 + 2^2 + (-5)^2} \cdot \sqrt{2^2 + (-3)^2 + 5^2}} \right|$$

$$\Rightarrow \cos \theta = \left| \frac{-25}{\sqrt{38} \cdot \sqrt{38}} \right|$$

$$\angle \theta = \cos^{-1} \left(\frac{25}{38} \right) \quad \underline{\underline{\text{Ans.}}}$$

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problem-6: Show that the planes $3x-4y+5z=0$ and $2x-y-2z=0$ are perpendicular to each other. $\rightarrow$ For practice.

problem-7: Show that the planes $6x-8y+10z=0$ and $3x-4y+5z=7$ are parallel to each other. $\rightarrow$ For practice.

problem-8: Find the equation of a plane through $(1, 2, 3)$ and parallel to plane $3x+4y-5z=0$.

solution: Let, the parallel plane $3x+4y-5z=0$ is $3x+4y-5z+k=0 \rightarrow (1)$

Equation (1) passing through $(1, 2, 3)$, we get

$$3 \cdot 1 + 4 \cdot 2 - 5 \cdot 3 + k = 0 \Rightarrow k = 4$$

putting $k=4$, from (1) $\Rightarrow$
 $3x+4y-5z+4=0$. Ans.

Problem-9: Show that equation of the plane through the point $(-1, 3, 2)$ and perpendicular to the planes $x+2y-2z=5$ and $3x+3y+2z=8$ is $10x-8y-3z+40=0$.

Solution: The equation of a plane which passes through a point $(-1, 3, 2)$ is $a(x+1)+b(y-3)+c(z-2)=0 \rightarrow$
where a, b, c are the direction ratios to the perpendicular plane. If two plane $x+2y-2z=5$ and $3x+3y+2z=8$ perpendicular to each.

$$\therefore a+2b-2c=0 \rightarrow (2)$$

$$\text{and } 3a+3b+2c=0 \rightarrow (3)$$

Eliminating a, b, c , from (1), (2), (3), we get

$$\begin{vmatrix} x+1 & y-3 & z-2 \\ 1 & 2 & -2 \\ 3 & 3 & 2 \end{vmatrix} = 0$$

$$\Rightarrow (x+1)(4+6) - (y-3)(2+6) + (z-2)(3-6) = 0$$

$$\Rightarrow 10(x+1) - 8(y-3) - 3(z-2) = 0$$

$$\Rightarrow 10x - 8y - 3z + 40 = 0 \text{ (shown)}.$$

Problem-10: Find the equation of the plane which passes through the points $(2, 2, 1)$ and $(9, 3, 6)$ and is perpendicular to the plane $2x + 6y + 6z + 9 = 0$.

Solution: The equation of a plane which passes through a point $(2, 2, 1)$ is

$$a(x-2) + b(y-2) + c(z-1) = 0 \rightarrow (1)$$

Equation (1) passes through the point $(9, 3, 6)$, we get, $a(9-2) + b(3-2) + c(6-1) = 0$

$$\Rightarrow 7a + b + 5c = 0 \rightarrow (2)$$

Where a, b, c are direction ratios to the perpendicular plane.

Given that, $2x + 6y + 6z + 9 = 0 \rightarrow (3)$

Equation (3) written as, $2a + 6b + 6c = 0 \rightarrow (4)$

Eliminating ^{a, b, c , from} (1), (2), (4), we get

$$\begin{vmatrix} x-2 & y-2 & z-1 \\ 7 & 1 & 5 \\ 2 & 6 & 6 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(6-30) - (y-2)(42-10) + (z-1)(42-2) = 0$$

$$\Rightarrow 24(x-2) + 32(y-2) - 40(z-1) = 0$$

$$\Rightarrow 3x + 4y - 5z = 9. \underline{\underline{Ans.}}$$

other method: The equation of any plane which passes through the points $(2, 2, 1)$ and $(9, 3, 6)$ is given by,

$$\begin{aligned} (x-x_1)(y-y_2) - (y-y_1)(x-x_2) &= A \{ (x-x_1)(z-z_2) \\ &- (x-x_2)(z-z_1) \} \\ \Rightarrow (x-2)(y-3) - (y-2)(x-9) &= A \{ (x-2)(z-6) \\ &- (x-9)(z-1) \} \\ \Rightarrow xy - 3x - 2y + 6 - xy + 9y + 2x - 18 &= A(xz - 6x - 2z \\ &+ 12 - xz + x + 9z - 9) \\ \Rightarrow -x + 7y - 12 &= A(-5x + 7z + 3) \\ \Rightarrow (5A-1)x + 7y - 7Az + (-3A-12) &= 0 \rightarrow (1) \end{aligned}$$

The plane (1) is perpendicular to the given plane
 $2x + 6y + 6z + 9 = 0$

$$\therefore 2 \cdot (5A-1) + 6 \cdot 7 + 6 \cdot (-7A) = 0$$

$$\Rightarrow -32A = -40$$

$$\Rightarrow A = \frac{5}{4}$$

putting the value of A in (1), we get

$$\Rightarrow (5 \cdot \frac{5}{4} - 1)x + 7y - 7 \cdot \frac{5}{4}z + (-3 \cdot \frac{5}{4} - 12) = 0$$

$$\Rightarrow 21x + 28y - 35z - 63 = 0$$

$$\therefore 3x + 4y - 5z - 9 = 0. \quad \underline{\text{(Ans)}}$$

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Problem-11: Find the equation of the plane passing through the intersection line of the planes $x-y+2z-3=0$ and $2x-y-3z=0$ and the point $(4, -3, 2)$.

Solution: The equation of the plane passing through the intersection line of the planes $(x-y+2z-3)+k(2x-y-3z)=0 \rightarrow (1)$
Equation (1) passes through the point $(4, -3, 2)$.

$$\therefore (4+3+2 \cdot 2-3)+k(2 \cdot 4+3-3 \cdot 2)=0$$

$$\Rightarrow 8+5k=0$$

$$\Rightarrow k = -8/5$$

Putting $k = -8/5$, from (1) $\Rightarrow$

$$(x-y+2z-3) - 8/5(2x-y-3z) = 0$$

$$\Rightarrow 5(x-y+2z-3) - 8(2x-y-3z) = 0$$

$$\Rightarrow 11x - 3y - 34z + 15 = 0. \quad \underline{\underline{Ans}}$$

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Problem-12: Find the equation of the plane which passes through the points $(1, 0, -1)$ and $(2, 1, 3)$ and is perpendicular to the plane $2x+y+z+1=0 \rightarrow$ For practice

Problem-13: Find the equation of a plane passing through the line of intersection of the planes $7x - 4y + 7z + 16 = 0$ and $4x + 3y - 2z + 3 = 0$ and is parallel to the plane $3x - 7y + 9z + 5 = 0$.

Solution: The equation of any plane which passes through the line of intersection of the two planes $7x - 4y + 7z + 16 = 0$ and $4x + 3y - 2z + 3 = 0$ is

$$(7x - 4y + 7z + 16) + k(4x + 3y - 2z + 3) = 0 \quad | \text{ } k \text{ is Constant}$$
$$\Rightarrow (7+4k)x + (3k-4)y + (7-2k)z + (3k+16) = 0 \rightarrow (1)$$

Since the plane (1) is parallel to the given plane

$$3x - 7y + 9z + 5 = 0$$

$$\therefore \frac{7+4k}{3} = \frac{3k-4}{-7} = \frac{7-2k}{9}$$

$$\text{Now, } 9k - 12 = -49 - 28k$$

$$\Rightarrow 37k = -37$$

$$\therefore k = -1$$

Putting the value of k in (1), we get

$$\Rightarrow 3x - 7y + 9z + 13 = 0$$

Which is the required equation.

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Problem-14: Find the equation of a plane passing through the line of intersection of the planes $x-2y+3z+4=0$ and $2x-3y+4z-1=0$ and is perpendicular to the plane $3x-y+2z-1=0$.

Solution: The equation of any plane which passes through the line of intersection of the two planes $x-2y+3z+4=0$ and $2x-3y+4z-1=0$ is

$$(x-2y+3z+4) + k(2x-3y+4z-1) = 0 \quad / \text{ } k \text{ is Constant}$$

$$\Rightarrow (1+2k)x + (-2-3k)y + (3+4k)z + (4-k) = 0 \rightarrow (1)$$

Since the plane (1) is perpendicular to the given plane $3x-y+2z-1=0$

$$\therefore 3(1+2k) - 1(-2-3k) + 2(3+4k) = 0$$

$$\Rightarrow 17k + 11 = 0$$

$$\Rightarrow k = -\frac{11}{17}$$

Putting the value k in (1), we get

$$\Rightarrow \left(1 - \frac{22}{17}\right)x + \left(-2 + \frac{33}{17}\right)y + \left(3 - \frac{44}{17}\right)z + \left(4 + \frac{11}{17}\right) = 0$$

$$\Rightarrow -5x - y + 7z + 79 = 0$$

$$\Rightarrow 5x + y - 7z - 79 = 0.$$

Which is the required equation of the plane.

problem-15: Find the equation of the plane which is perpendicular to the plane $5x + 3y + 6z + 8 = 0$ and contains the line of intersection of the planes $x + 2y + 3z - 4 = 0$ and $2x + y - z + 5 = 0$ $\rightarrow$ For practice

problem-16: Find the equation of the plane that passing through the point $(2, 1, 4)$ and perpendicular to both the planes $x + y - z = 0$ and $9x - 7y + 6z + 48 = 0$.

Solution: The equation of any plane passing through the point $(2, 1, 4)$ is

$$a(x-2) + b(y-1) + c(z-4) = 0 \rightarrow (1)$$

The plane (1) is perpendicular to each of the planes $x + y - z = 0$ and $9x - 7y + 6z + 48 = 0$

$$\therefore a + b - c = 0$$

$$\text{and } 9a - 7b + 6c = 0$$

$$\text{So, } \frac{a}{6-7} = \frac{b}{-9-6} = \frac{c}{-7-16}$$

$$\Rightarrow \frac{a}{1} = \frac{b}{15} = \frac{c}{16} = k \text{ (say)}$$

$$\therefore a = k, b = 15k, c = 16k$$

Putting the values of a, b, c in (1), we get

$$k(x-2) + 15k(y-1) + 16k(z-4) = 0$$

$\Rightarrow x + 15y + 16z - 81 = 0$. Which is the required equation.

Problem-17: Find the equation of the plane that passes through the point $(2, -3, 1)$ and is normal to the line joining the points $(3, 4, -1)$ and $(2, -1, 5)$.

Solution: The equation of any plane passing through the point $(2, -3, 1)$ is

$$a(x-2) + b(y+3) + c(z-1) = 0 \longrightarrow (1)$$

The direction ratios of the line joining the two points $(3, 4, -1)$ and $(2, -1, 5)$ are

$$2-3, -1-4, 5+1$$

$$\text{i.e., } 1, 5, -6$$

Equation (1) is normal to the line joining the points, we get

$$\frac{a}{1} = \frac{b}{5} = \frac{c}{-6} = k \text{ (say)}$$

$$\Rightarrow a = k, b = 5k, c = -6k$$

Putting the values of a, b, c in (1), we get

$$k(x-2) + 5k(y+3) - 6k(z-1) = 0$$

$$\Rightarrow x + 5y - 6z + 19 = 0.$$

Which is the required equation.

Problem - 18: Find the equation of the planes through the line of intersection of the planes $2x + y + 2z + 3 = 0$ and $x + 2y - z = 0$ and distance 1 unit from the origin.

Solution: The equation of the planes through the line of intersection of the planes $2x + y + 2z + 3 = 0$ and $x + 2y - z = 0$ is

$$(2x + y + 2z + 3) + k(x + 2y - z) = 0$$
$$\Rightarrow (2+k)x + (1+2k)y + (2-k)z + 3 = 0 \quad (1)$$

Distance 1 unit from the origin of (1), we get

$$\Rightarrow \frac{3}{\sqrt{(2+k)^2 + (1+2k)^2 + (2-k)^2}} = \pm 1$$

$$\Rightarrow \frac{3}{\sqrt{9 + 4k + 6k^2}} = \pm 1$$

$$\Rightarrow 9 = 9 + 4k + 6k^2$$

$$\Rightarrow k(6k + 4) = 0$$

$$\therefore k = 0, -\frac{2}{3}$$

Putting $k = 0, -\frac{2}{3}$ in (1), we get

$$2x + y + 2z + 3 = 0 \text{ and } 4x - y + 8z + 9 = 0$$

Which is the required equation of planes.

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